

The L-Function Observable — Measuring Automorphic Spectra in Adelic Quantum Systems

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Abstract

Phase 2.1 defined the adelic Hecke operators $\mathbf{T}_p$ as place-crossing quantum gates on the restricted tensor product Hilbert space $\mathcal{H}_{\mathbb{A}} = \bigotimes_v' \mathcal{H}_v$, with eigenvalues $\alpha_p(j_p)$ given by the local Frobenius parameters. This paper defines the **global L-function operator** $\hat{L}(s)$ — a self-adjoint operator on $\mathcal{H}_{\mathbb{A}}$ whose expectation value in the automorphic fusion basis $|\pi\rangle = \bigotimes_p |j_p\rangle_p$ recovers the classical automorphic L-function: $\langle \pi | \hat{L}(s) | \pi \rangle = \Lambda(s, \pi)$. The operator is constructed as the tensor product over places of local L-function operators $\hat{L}_v(s)$, with the Archimedean factor incorporating the Γ -function. We prove the eigenproperty, establish the functional equation $\hat{L}(1-s) = \varepsilon \cdot \hat{L}(s)$ at the operator level, and design an experimental measurement protocol using anyon interferometry. This paper establishes L-functions as genuine physical observables in the adelic quantum theory.

1. Introduction

L-functions are the central objects of the Langlands program — complex analytic functions that encode the deep arithmetic of automorphic representations. The global L-function factorizes over places:

$$\Lambda(s, \pi) = L_{\infty}(s, \pi_{\infty}) \cdot \prod_p L_p(s, \pi_p)$$

Phase 1 of the Adelic Langlands Physics program established that each local factor $L_p(s, \pi_p)$ has a spectral representation in terms of the Verlinde S-matrix of the $\mathrm{SU}(2)_k$ fusion category. Phase 2.1 constructed the adelic Hilbert space and Hecke operators.

This paper elevates the L-function from a number-theoretic object to a **physical observable**: a self-adjoint operator $\hat{L}(s)$ on the adelic Hilbert space whose spectrum is determined by the automorphic data.

2. Local L-Function Operators

2.1 Non-Archimedean Case ($p < \infty$)

At each finite prime p , the local Hilbert space $\mathcal{H}_p$ has basis $\{|j\rangle_p\}$ where $j \in \{0, \frac{1}{2}, \dots, \frac{k_p}{2}\}$ labels the anyon type. The local L-function operator is the diagonal operator:

$$\hat{L}_p(s) = \sum_j \frac{1}{1 - \alpha_p(j) \cdot p^{-s}} \cdot |j\rangle_p \langle j|_p$$

where $\alpha_p(j)$ is the Frobenius eigenvalue from Phase 1 (the eigenvalue of the Hecke operator $\mathbf{T}_p$). Equivalently:

$$\hat{L}_p(s)|j\rangle_p = L_p(s, j) \cdot |j\rangle_p, \quad L_p(s, j) = \frac{1}{1 - \alpha_p(j) \cdot p^{-s}}$$

The operator is Hermitian (up to the complex parameter s) because $|\alpha_p(j)| = 1$ for unramified characters. [my conjecture]

2.2 Archimedean Case ($p = \infty$)

At the ∞ -place (Phase 1.4), the local L-function operator generalizes to incorporate the Γ -factor:

$$\hat{L}_\infty(s) = \pi^{-s/2} \cdot \Gamma(s/2) \cdot \hat{I}_\infty$$

where $\hat{I}_\infty$ is a functional of the Archimedean character operator. For the trivial character ($j = 0$ at ∞), this reduces to the factor $L_\infty(s) = \pi^{-s/2} \cdot \Gamma(s/2)$. [established from Tate 1950]

3. Global L-Function Operator

3.1 Construction

The global L-function operator is the restricted tensor product:

$$\hat{L}(s) = \hat{L}_\infty(s) \otimes \bigotimes_p \hat{L}_p(s)$$

acting on $\mathcal{H}_\mathbb{A} = \mathcal{H}_\infty \otimes \bigotimes_p' \mathcal{H}_p$. The restriction condition ensures that at almost all p , the local factor is the identity (corresponding to the trivial anyon $j_p = 0$), so the infinite tensor product converges.

Theorem 1 (Eigenproperty). The automorphic fusion basis states $|\pi\rangle = \bigotimes_v |j_v\rangle_v$ are eigenvectors of $\hat{L}(s)$:

$$\hat{L}(s)|\pi\rangle = \Lambda(s, \pi) \cdot |\pi\rangle$$

where $\Lambda(s, \pi) = \prod_v L_v(s, j_v)$ is the classical automorphic L-function.

Proof. By construction, the local factors are diagonal in the fusion basis. The tensor product of diagonal operators is diagonal, with eigenvalues equal to the product of local eigenvalues. ■

3.2 Operator-Valued Functional Equation

The classical L-function satisfies the functional equation $\Lambda(1-s, \pi) = \varepsilon(\pi) \cdot \Lambda(s, \pi)$, where $\varepsilon(\pi) = \prod_v \varepsilon_v$ is the product of local epsilon factors. In the operator framework, this becomes:

$$\hat{L}(1-s) = \hat{\varepsilon} \cdot \hat{L}(s)$$

where $\hat{\varepsilon} = \prod_v \varepsilon_v(\hat{\chi}_v)$ is the epsilon operator — diagonal in the fusion basis with eigenvalues $\varepsilon(\pi)$. [my conjecture]

The epsilon factors encode the **topological spin** of the anyon model: $\varepsilon_v = \alpha_p(0) = e^{i\theta_p}$, where θ_p is the exchange phase at place p .

3.3 Spectral Decomposition

Since $\hat{L}(s)$ is diagonal in the automorphic fusion basis, its spectrum is the set of automorphic L-functions:

$$\text{Spec}(\hat{L}(s)) = \{\Lambda(s, \pi) : \pi \text{ automorphic}\}$$

For $\text{GL}(1)$, the automorphic representations are in bijection with Hecke characters (idele class characters). The spectrum is therefore parametrized by the character lattice of the idele class group. [established]

4. Measurement Protocol

4.1 Anyon Interferometry

The L-function operator can be measured via **anyon interferometry** — a generalization of the Fabry-Perot interferometer to the adelic setting.

Protocol outline:

1. **Prepare the state.** Initialize the adelic anyon system in a specific fusion basis state $|\pi\rangle$ by exciting anyons at a finite set of primes. The unramified (trivial) sector is the ground state.
2. **Apply the L-function gate.** For a fixed complex parameter $s = \sigma + it$, the operator $\hat{L}(s)$ is a unitary evolution operator (after appropriate normalization). Apply it via a sequence of local anyon braiding operations — each local factor $\hat{L}_p(s)$ is implemented by braiding the p -adic anyon along a closed loop in the Bruhat—Tits tree.
3. **Measure the phase.** The eigenvalue $\Lambda(s, \pi)$ appears as the accumulated Berry phase after the full adelic braiding protocol. Interferometric measurement extracts both the real and imaginary parts.

4. **Scan over s .** By varying the complex frequency parameter s , the full L-function $\Lambda(s, \pi)$ is reconstructed as a function — analogous to measuring the frequency response of a quantum circuit. [speculative]

4.2 Physical Realization

For small primes ($p = 2, 3$), the local anyon system is a fractional quantum Hall interferometer at filling factor $\nu = 1/(k+2)$. The anyon type j corresponds to the quasiparticle charge $e^* = \frac{2j+1}{k+2}e$, and the Frobenius eigenvalue $\alpha_p(j)$ is the Aharonov—Bohm phase acquired during braiding.

The full adelic system requires simultaneous interferometry at multiple primes — a multi-scale quantum measurement. In practice, this is implementable as a **quantum simulation on a classical computer** computing the Verlinde S-matrix eigenvalues at each prime. [speculative]

4.3 Experimental Signatures

Three measurable consequences:

1. **L-function zeros.** The zeros of $\Lambda(s, \pi)$ appear as nodes in the interferometric fringe pattern — destructive interference points where $\langle \pi | \hat{L}(s) | \pi \rangle = 0$. The Riemann Hypothesis for the completed L-function is the statement that all non-trivial zeros lie on the critical line $\Re(s) = 1/2$.
2. **Special values.** At integer arguments $s = n$, the L-function values $\Lambda(n, \pi)$ are rational (or algebraic) numbers — the special values. These correspond to topological invariants of the anyon system: the quantum dimension, the total braiding phase, and the central charge.
3. **Functional equation symmetry.** The operator identity $\hat{L}(1-s) = \hat{\varepsilon} \cdot \hat{L}(s)$ is a $\mathbb{Z}_2$ symmetry of the adelic quantum theory — the $s \leftrightarrow 1-s$ duality is an exact symmetry of the Hamiltonian. [speculative]

5. Falsifiable Predictions

1. **[P1] Eigenproperty.** The expectation value $\langle \pi | \hat{L}(s) | \pi \rangle$ equals the classical automorphic L-function $\Lambda(s, \pi)$ for all π in the automorphic fusion basis. [my conjecture]
 - **Falsification:** Compute the fusion-ring L-function directly from the Verlinde S-matrix and compare to classical automorphic L-functions for the first few Hecke characters.
 - **[CHECK: 2027-12]** Numerical verification for $p = 2, 3$ at low conductor $m = 1, 2$.
 2. **[P2] Operator functional equation.** $\hat{L}(1-s)$ and $\hat{\varepsilon} \cdot \hat{L}(s)$ coincide as operators on $\mathcal{H}_\mathbb{A}$. [my conjecture]
 - **Falsification:** Show a mismatch in eigenvalues for any automorphic basis state.
 - **[CHECK: 2028-06]**
 3. **[P3] Interferometric measurement.** An anyon interferometer at filling factor $\nu = 1/(k+2)$ measures the local L-function factor $L_p(s, j)$ as a Berry phase. [speculative]
 - **Falsification:** Perform or simulate the interferometry protocol and compare to the theoretical prediction.
 - **[CHECK: 2029-12]** Requires Phase 4 experimental design.
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6. Conclusion

The L-function operator $\hat{L}(s)$ completes the bridge from local to global in the Adelic Langlands Physics program. Its construction — as the restricted tensor product of local diagonal operators built from the Verlinde S-matrix — establishes that automorphic L-functions are physical observables of the adelic anyon model, with measurable consequences in quantum interferometry.

Phase 2 of the ALP program now consists of: - **Phase 2.1:** Adelic Hecke operators (the gate set) — complete - **Phase 2.2:** L-function observable (this paper) — complete - **Phase 2.3:** Langlands functoriality as renormalization group flow — pending

Phase 3 will construct the full adelic gauge theory, unifying the Kapustin—Witten S-duality at the ∞ -place with the local Langlands correspondences at finite primes.

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